

PLL FM Transmitter

JOY MUKHERJI

This is a circuit that offers a challenge to electronics enthusiasts and hobbyists—an FM transmitter that uses readily available components and covers the FM broadcast

band in 100kHz steps. The frequency drift is controlled by phase-locked-loop



(PLL) method, making transmission frequency rock-solid all the time, just like commercial stations. Using DIP switches, simply punch in the frequency on which you wish to broadcast, and you are on the air.

Circuit and working

The circuit comprises two units. Unit 1 is the RF section of the transmitter, shown in Fig. 1, which transmits the FM signal. Unit 2 is the PLL control section, shown in Fig. 2, which helps in locking the transmission frequency. The circuit uses phase-lock loop that provides drift-free transmission frequency.

The RF section is built around transistors T1-T4, with T1 (2N2222) in Colpitts oscillator configuration. The frequency of the oscillator is determined by coil L1 and capacitors C1, C2, C3 and C4. Modulating signal, which is in audio range, is fed through Jack1.

D1 is a varactor diode, working in reverse-bias mode. Since this is an FM transmitter, the deviation in the frequency of the oscillator is based on the amount of reverse-bias generated by the audio signal. Transistor T2 (2N2222) acts as a buffer that isolates the oscillator from the rest of the amplifier chain.

Frequency-modulated signal is coupled to driver transistor T3 (BF199) via capacitor C5. R7 is a current-limiting resistor. Transistor T3 is wired as a Class A amplifier and drives transistor T4 (2N3866A) via broadband impedance matching transformer L2. Power amplifier operates in Class B mode. Coil L4 and trimmer capacitor VC1 match transistor T4's collector to the antenna.

The PLL control section is built around 64/256 prescaler SAB6456A (IC1), programmable divide-by-N counter CD4059 (IC3), phase locked-loop comparator CD4046 (IC4), 14-stage ripple carry binary counter/oscillator

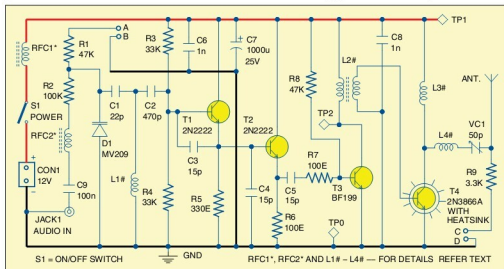


Fig. 1: RF section of the transmitter (Unit 1)

PARTS LIST

Semiconductors:

IC1	- SAB6456A 64/256 prescaler
IC2	- 7805 voltage regulator
IC3	- CD4059 programmable divide-by-N counter
IC4	- CD4046 phase-locked-loop (PLL) comparator
IC5	- CD4060 ripple carry binary counter/oscillator
IC6	- TL071 operational amplifier
T1, T2, T6	- 2N2222 npn transistor
T3, T5	- BF199 npn transistor
T4	- 2N3866A npn transistor
D1	- MV209 varactor diode
LED1	- 5mm LED

Resistors (all 1/4-watt, ±5% carbon):

R1, R8, R19-R31	- 47-kilo-ohm
R2, R15	- 100-kilo-ohm
R3, R4	- 33-kilo-ohm
R5	- 330-ohm
R6, R7	- 100-ohm
R9	- 3.3-kilo-ohm
R10	- 18-kilo-ohm
R11, R12, R16	- 10-kilo-ohm
R13	- 220-ohm
R14	- 2.2-kilo-ohm
R17	- 4.7-kilo-ohm
R18	- 1-mega-ohm
VR1	- 100-kilo-ohm preset

Capacitors:

C1, C16, C17	- 22pF, ceramic disk
--------------	----------------------

C2	- 470pF ceramic disk
C3-C5	- 15pF ceramic disk
C6, C8	- 1nF ceramic disk
C7, C21	- 1000µF, 25V electrolytic
C9, C14, C18-C20, C22, C24	- 100nF ceramic disk
C10, C15	- 10µF, 25V electrolytic
C11	- 200nF ceramic disk
C12	- 400nF ceramic disk
C13	- 220µF, 25V electrolytic
VC1	- 50pF trimmer capacitor

Miscellaneous:

RFC1	- 50T, 28SWG balun core
RFC2	- 25T, 28SWG balun core
L1	- 4T, 4mm dia 26SWG air core
L2	- Broadband transformer 6T primary 26SWG, 1T secondary, 20SWG
L3	- 13T, 8mm dia 26SWG air core
L4	- 6T, 8mm dia 26SWG air core
CON-CON2	- 2-pin terminal connector
X ₁ , 1	- 6.4MHz crystal
DIP1-DIP4	- DIP switch
S1, S2	- On/off switch
	- Audio input jack
	- Co-axial cable
	- Heat sink for transistor T4
	- Wire antenna